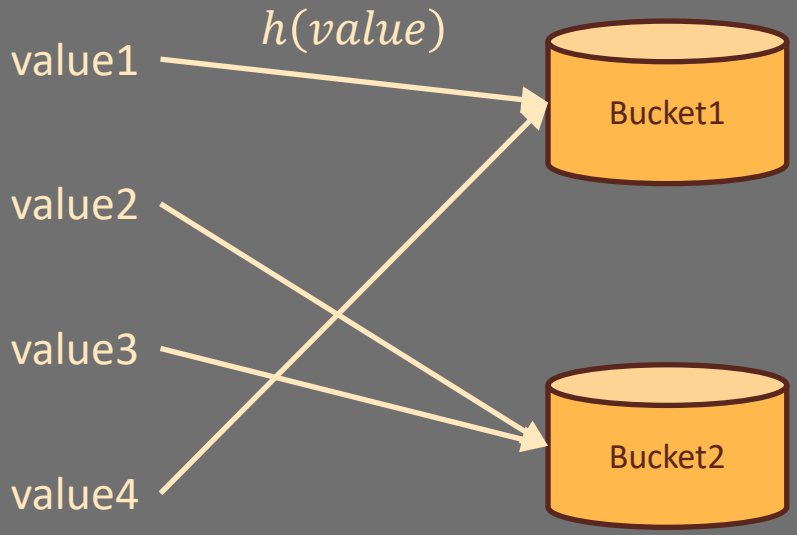
Hash Indices

An alternative to ordered indices

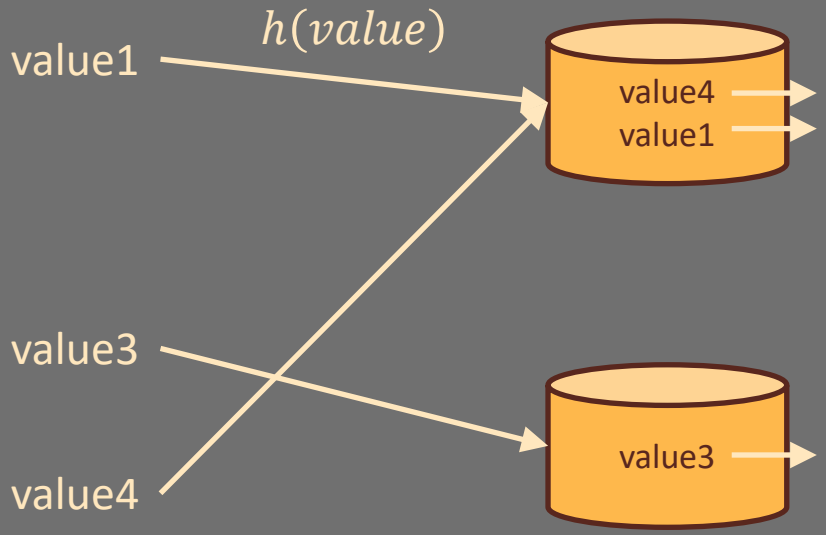
Basic idea of a hash function



$h(\cdot)$ distributes values uniformly.

$h(\cdot)$ distributes typical subsets of values uniformly.

Using hash function for indexing



1. Use $h(\cdot)$ to find bucket.
2. Search/insert/delete from bucket. Bucket items possibly sorted.

Motivation: $O(1)$

Buckets can overflow

- Overflow some – skewed usage
- Overflow many – not enough space reserved

Techniques to overcome:

- Chain additional buckets – degrades to linear search
- Stop and reorganize
- Dynamic hashing (extensible or linear hashing) – techniques that allow the number of buckets to grow without rehashing existing data

Trade-offs

Hash

- Space:
  - Performs best if buckets half-empty.
- Time:
  - $O(1)$
  - Degrades to linear access.
- Operations:
  - Can't search a range.

B+-tree

- Space:
  - Pointers & internal nodes
- Time:
  - $O(\log n)$
  - Doesn't degrade.
- Operations:
  - Can search a range.